Dense cropping attracted pests, which had to be dealt with the increasing use of toxic pesticides. Over time, the natural fertility of the soil has declined, necessitating progressively larger infusions of fertilisers and water. The water table in most areas has fallen precipitously, creating acute water scarcity. The use of chemical fertilisers and pesticides has led to the contamination of water supplies with serious effects on human and animal health. Farmers using such toxic chemicals hardly ever use protective clothing or shields and fall prey to a variety of skin, respiratory and eye diseases. Many farming families have been bankrupted by huge health costs rather than crop failure. Subsidies for fertilisers and pesticides only compound the crisis.

Despite proven and sustainable alternatives being available, Indian agriculture is locked into a pattern of production that has long outlived its utility and is poisoning our water sources, burning our soils and bringing death and disease to our rural population, even while making agriculture a significant source of greenhouse gas emissions. A new and sustainable strategy of agricultural development focused on making the farmer a viable economic agent rather than targeting only crop yields is the need of the hour. Successful experiments, for example, from Andhra Pradesh, could be scaled up but then this means confronting the powerful fertiliser and pesticide lobbies in the country. India's experience in this critical field could become a model for many countries in Asia and Africa that suffer from similar challenges.

In aspiring to the role of a global model, India should take the lead on climate and ecology in the Indian subcontinent. As the largest, most capable country in its region, India can help evolve a South Asian Climate Change Action Plan and an Ecological Integrity Plan. The melting of glaciers in the Himalayas is advancing relentlessly and, if unchecked, could turn the subcontinent's perennial rivers into seasonal rainfed streams. Sea-level rise due to global warming will not only pose an existential threat to the island countries of Sri Lanka and the Maldives, but also threaten low-lying deltaic plains in Bangladesh, India and Pakistan. The rivers of the subcontinent are living systems and cannot be fragmented into national flows as if they were pieces of private property. Only cross-national river basin management can work. Our region is one of the biodiversity hotspots of the world, with a diversity of flora and fauna unmatched anywhere in the world. This rich biodiversity is being degraded through the wanton cutting down of forests and the encroachment of human populations into the habitats of wild species. We need cross-border forest corridors for the surviving wild animals for which regional cooperation is indispensable. India needs to work together with all its South Asian neighbours to make the whole of South Asia a model of sustainable development. This would lend credibility to India's aspirations to be its global model. This is quite apart from the fact that in an ecologically integrated space like South Asia, national solutions alone will not work. Our rivers, our mountains, our forests and our wildlife, all constitute our common, inherited and shared space. Only through regional collaboration can these be managed sustainably. LiFE needs to become regional before it goes global.

AS THE MOST CAPABLE COUNTRY IN THE REGION, INDIA CAN HELP EVOLVE A **SOUTH ASIAN CLIMATE CHANGE** PLAN AND AN **ECOLOGICAL INTEGRITY** PLAN

India is the current chair of the G-20, a grouping of the world's most influential countries. The summit in 2023, under India's chairmanship, could provide the platform for addressing the ecological crisis that threatens our planet's survival. The year gone by saw two lost opportunities for global collaborative action to address the challenge of climate and the associated but larger challenge of diminishing biodiversity. The outcome of the 27th Conference of Parties to the UN Framework Convention on Climate Change (COP-27) held in Sharm El-Sheikh in Egypt in November and that of the 15th Conference of Parties to the Biodiversity Convention (COP-15) convened in Montreal, failed to live up to international expectations of urgent action to address the global ecological crisis of which climate change and biodiversity loss are twin manifestations.

On this count, the UN has declared: "The ocean generates 50 per cent of the oxygen we need, absorbs 25 per cent of all carbon dioxide emissions and captures 90 per cent of the excess heat generated by these emissions. It is not just the lungs of the planet but also its largest carbon sink—a vital buffer against the impacts of climate change."

With the dumping of plastics and hazardous waste into the oceans, their acidification and thermal expansion due to global warming, the ocean's role as a carbon sink is declining. Many other such feedback loops demand a comprehensive approach to the ecological crisis perhaps through an International Ecology Convention. India could, as G-20 chair, pioneer the approach. India should lead by the power of its example, not just through exhortation. ■